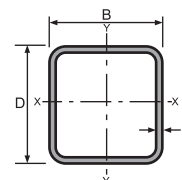


# HOT FORMED HOLLOW SECTIONS



## SQUARE

| Section size<br>D x B | Thickness<br>t | Unit weight<br>M | Section area<br>A | Moment of inertia<br>I | Radius of gyration<br>r | Elastic modulus<br>Z | Plastic modulus<br>S | Torsional<br>J  | Constants<br>C  | Superficial area per m | Approx. length per tonne |
|-----------------------|----------------|------------------|-------------------|------------------------|-------------------------|----------------------|----------------------|-----------------|-----------------|------------------------|--------------------------|
| mm                    | mm             | kg/m             | cm <sup>2</sup>   | cm <sup>4</sup>        | cm                      | cm <sup>3</sup>      | cm <sup>3</sup>      | cm <sup>4</sup> | cm <sup>3</sup> | m <sup>2</sup>         | m                        |
| 20 x 20               | 2.0            | 1.12             | 1.42              | 0.76                   | 0.73                    | 0.76                 | 0.95                 | 1.22            | 1.07            | 0.076                  | 893                      |
|                       | 2.5*           | 1.35             | 1.72              | 0.87                   | 0.71                    | 0.88                 | 1.12                 | 1.41            | 1.21            | 0.075                  | 741                      |
| 25 x 25               | 2.0*           | 1.43             | 1.82              | 1.59                   | 0.94                    | 1.27                 | 1.56                 | 2.52            | 1.81            | 0.096                  | 700                      |
|                       | 2.5*           | 1.74             | 2.22              | 1.85                   | 0.91                    | 1.48                 | 1.86                 | 2.97            | 2.09            | 0.095                  | 574                      |
|                       | 3.0*           | 2.04             | 2.60              | 2.06                   | 0.89                    | 1.65                 | 2.12                 | 3.36            | 2.31            | 0.094                  | 490                      |
|                       | 3.2*           | 2.15             | 2.74              | 2.14                   | 0.88                    | 1.71                 | 2.21                 | 3.49            | 2.38            | 0.093                  | 466                      |
| 30 x 30               | 2.5*           | 2.14             | 2.72              | 3.40                   | 1.12                    | 2.27                 | 2.79                 | 5.40            | 3.22            | 0.115                  | 469                      |
|                       | 3.0*           | 2.51             | 3.20              | 3.84                   | 1.10                    | 2.56                 | 3.21                 | 6.17            | 3.61            | 0.114                  | 399                      |
|                       | 3.2            | 2.65             | 3.38              | 4.00                   | 1.09                    | 2.67                 | 3.37                 | 6.45            | 3.75            | 0.113                  | 378                      |
| 40 x 40               | 2.5*           | 2.92             | 3.72              | 8.67                   | 1.53                    | 4.33                 | 5.21                 | 13.6            | 6.23            | 0.155                  | 343                      |
|                       | 3.0*           | 3.45             | 4.40              | 9.96                   | 1.51                    | 4.98                 | 6.07                 | 15.7            | 7.11            | 0.154                  | 290                      |
|                       | 3.2            | 3.66             | 4.66              | 10.4                   | 1.50                    | 5.22                 | 6.40                 | 16.5            | 7.43            | 0.153                  | 274                      |
|                       | 4.0            | 4.46             | 5.68              | 12.1                   | 1.46                    | 6.07                 | 7.61                 | 19.5            | 8.56            | 0.151                  | 225                      |
|                       | 5.0*           | 5.40             | 6.88              | 13.8                   | 1.42                    | 6.92                 | 8.92                 | 22.6            | 9.65            | 0.149                  | 185                      |
| 50 x 50               | 2.5*           | 3.71             | 4.72              | 17.7                   | 1.94                    | 7.07                 | 8.38                 | 27.4            | 10.2            | 0.195                  | 270                      |
|                       | 3.0*           | 4.39             | 5.60              | 20.5                   | 1.91                    | 8.20                 | 9.83                 | 32.0            | 11.8            | 0.194                  | 228                      |
|                       | 3.2            | 4.66             | 5.94              | 21.6                   | 1.91                    | 8.62                 | 10.4                 | 33.8            | 12.4            | 0.193                  | 215                      |
|                       | 4.0            | 5.72             | 7.28              | 25.5                   | 1.87                    | 10.2                 | 12.5                 | 40.4            | 14.5            | 0.191                  | 175                      |
|                       | 5.0            | 6.97             | 8.88              | 29.6                   | 1.83                    | 11.9                 | 14.9                 | 47.6            | 16.7            | 0.189                  | 144                      |
|                       | 6.0            | 8.15             | 10.4              | 33.0                   | 1.78                    | 13.2                 | 17.0                 | 53.7            | 18.5            | 0.187                  | 123                      |
|                       | 6.3*           | 8.49             | 10.8              | 33.9                   | 1.77                    | 13.6                 | 17.5                 | 55.3            | 18.9            | 0.186                  | 118                      |
| 60 x 60               | 3.0*           | 5.34             | 6.80              | 36.6                   | 2.32                    | 12.2                 | 14.5                 | 56.9            | 17.7            | 0.234                  | 188                      |
|                       | 3.2            | 5.67             | 7.22              | 38.7                   | 2.31                    | 12.9                 | 15.3                 | 60.1            | 18.6            | 0.233                  | 177                      |
|                       | 4.0            | 6.97             | 8.88              | 46.1                   | 2.28                    | 15.4                 | 18.6                 | 72.4            | 22.1            | 0.231                  | 144                      |
|                       | 5.0            | 8.54             | 10.9              | 54.4                   | 2.24                    | 18.1                 | 22.3                 | 86.3            | 25.8            | 0.229                  | 118                      |
|                       | 6.0            | 10.0             | 12.8              | 61.5                   | 2.19                    | 20.5                 | 25.7                 | 98.6            | 28.9            | 0.227                  | 100                      |
|                       | 6.3*           | 10.5             | 13.3              | 63.4                   | 2.18                    | 21.1                 | 26.6                 | 102             | 29.7            | 0.226                  | 95.3                     |
|                       | 8.0*           | 12.8             | 16.3              | 72.4                   | 2.11                    | 24.1                 | 31.4                 | 119             | 33.5            | 0.223                  | 78.1                     |
| 70 x 70               | 3.0*           | 6.28             | 8.0               | 59.6                   | 2.73                    | 17.0                 | 20.0                 | 92.1            | 24.8            | 0.274                  | 160                      |
|                       | 3.6            | 7.46             | 9.5               | 69.5                   | 2.70                    | 19.9                 | 23.6                 | 108             | 28.7            | 0.272                  | 135                      |
|                       | 5.0            | 10.1             | 12.9              | 90.1                   | 2.64                    | 25.7                 | 31.2                 | 142             | 36.8            | 0.269                  | 99.1                     |
|                       | 6.0            | 11.9             | 15.2              | 103                    | 2.60                    | 29.4                 | 36.2                 | 163             | 41.7            | 0.267                  | 84.0                     |
|                       | 6.3*           | 12.5             | 15.9              | 106                    | 2.59                    | 30.4                 | 37.6                 | 169             | 43.0            | 0.266                  | 80.0                     |
|                       | 8.0*           | 15.3             | 19.5              | 123                    | 2.51                    | 35.3                 | 45.0                 | 200             | 49.4            | 0.263                  | 65.4                     |
| 75 x 75               | 3.0            | 6.75             | 8.6               | 74.0                   | 2.93                    | 19.7                 | 23.2                 | 114             | 28.8            | 0.294                  | 148                      |
|                       | 3.6            | 8.02             | 10.2              | 86.5                   | 2.91                    | 23.1                 | 27.3                 | 134             | 33.4            | 0.292                  | 125                      |
|                       | 5.0            | 10.9             | 13.9              | 113                    | 2.85                    | 30.0                 | 36.3                 | 177             | 43.1            | 0.289                  | 91.7                     |
|                       | 6.0            | 12.9             | 16.4              | 129                    | 2.81                    | 34.4                 | 42.2                 | 204             | 49.0            | 0.287                  | 77.5                     |
|                       | 6.3            | 13.4             | 17.1              | 134                    | 2.79                    | 35.6                 | 43.8                 | 212             | 50.6            | 0.286                  | 74.6                     |
|                       | 8.0            | 16.6             | 21.1              | 156                    | 2.72                    | 41.7                 | 52.6                 | 252             | 58.6            | 0.283                  | 60.2                     |
| 80 x 80               | 3.0*           | 7.22             | 9.2               | 90.6                   | 3.14                    | 22.7                 | 26.5                 | 139             | 33.1            | 0.314                  | 139                      |
|                       | 3.6            | 8.59             | 10.9              | 106                    | 3.11                    | 26.5                 | 31.3                 | 164             | 38.5            | 0.312                  | 117                      |
|                       | 5.0            | 11.7             | 14.9              | 139                    | 3.05                    | 34.7                 | 41.7                 | 217             | 49.8            | 0.309                  | 85.5                     |
|                       | 6.0            | 13.8             | 17.6              | 159                    | 3.01                    | 39.8                 | 48.6                 | 251             | 56.9            | 0.307                  | 72.5                     |
|                       | 6.3            | 14.4             | 18.4              | 165                    | 3.00                    | 41.3                 | 50.5                 | 261             | 58.8            | 0.306                  | 69.5                     |
|                       | 8.0*           | 17.8             | 22.7              | 194                    | 2.92                    | 48.6                 | 60.9                 | 312             | 68.5            | 0.303                  | 56.2                     |

Notes:  
 Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.  
 \* Size not included in BS 4848 Part 2  
 † 43C only  
 Length per tonne rounded up